

Topos Structure Pt. 1

Topoi: The Categorical Analysis of Logic

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Recall

So far, we have been assembling the ingredients to discuss topoi:

- ▶ Categories
- ▶ Terminal objects
- ▶ Products
- ▶ Subobjects
- ▶ Pullbacks
- ▶ Subobject classifiers
- ▶ Characteristic morphisms
- ▶ Equalizers

Today, we finally get to the **logic** part!

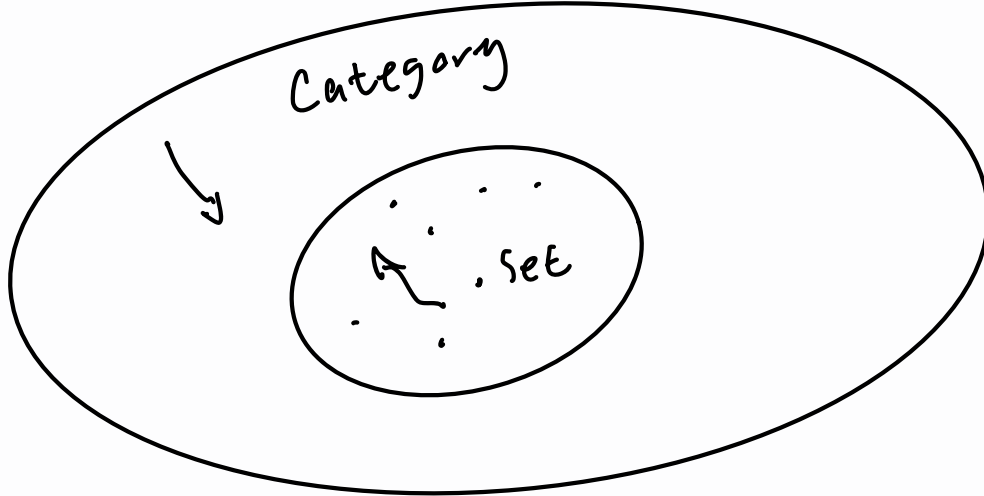
Goals for today

By the end of this meeting, we should understand:

- ▶ The definition of a **topos** and some **examples**
- ▶ What **sheaves** and **monoids** are, and how they relate to logic
- ▶ Some **set theory** analogues in topos theory
 - Subobjects
 - Power objects
 - Comprehension
- ▶ As time permits, **topos structure**
 - Monomorphisms in topoi are equalizers
 - Epi-mono Factorization
 - Other cool facts

Illustration

To define something "set-like", we have competing priorities.



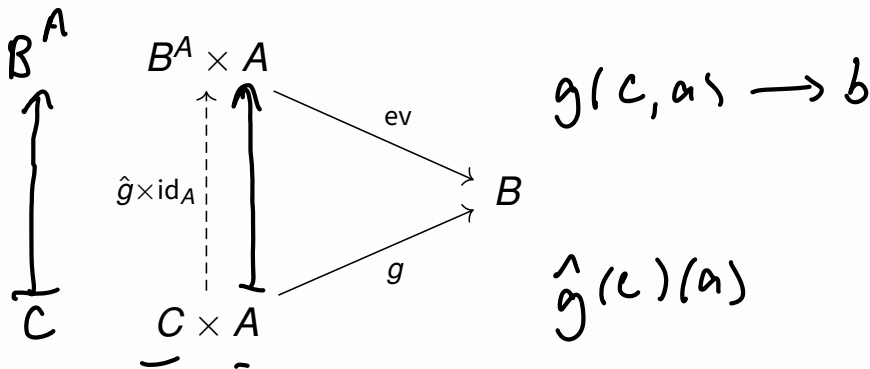
Prerequisites

Exponentiation in Set

Intuition

In **Set**, we have $B^A = \{f : A \rightarrow B\}$ that comes with a special morphism $\text{ev} : B^A \times A \rightarrow B$ such that $\text{ev}(\langle f, x \rangle) = f(x)$.

For the categorical description, let's examine all functions that look like $g : C \times A \rightarrow B$.



Exponentiation in General

Definition

A category $\mathcal{C}$ has **exponentiation** if for any objects a, b there is an object b^a and a morphism $\text{ev} : b^a \times a \rightarrow b$ such that for any c and morphism $g : c \times a \rightarrow b$, there is a unique $\hat{g}$ such that $\text{ev} \circ (\hat{g} \times a) = g$.

A commutative triangle diagram illustrating the universal property of exponentiation. The top vertex is labeled $b^a \times a$, the bottom-left vertex is labeled $c \times a$, and the right vertex is labeled b . A dashed vertical arrow points from $c \times a$ to $b^a \times a$ and is labeled $\hat{g} \times \text{id}_a$. A solid arrow points from $b^a \times a$ to b and is labeled ev . A solid arrow points from $c \times a$ to b and is labeled g .

Why we care

$$\mathcal{C}(a, b) = \text{Hom}_{\mathcal{C}}(a, b) = \{a \rightarrow b\}$$

This amounts to saying that $\mathcal{C}(c, b^a) \cong \mathcal{C}(c \times a, b)$, or that **the collection of morphisms** (or hom set) is **internal** to the category.

Intuition $\downarrow$ $1 \rightarrow b^a$ $a \rightarrow b$ $f: A \rightarrow B$

Take $c = 1$ and get $\mathcal{C}(1, b^a) \cong \mathcal{C}(a, b)$. $\mathcal{C}(1, b^a)$ are the **global elements** of b^a which are exactly the collection of morphisms from $a \rightarrow b$.

Yet, we have a stronger case where we can probe by *any* object C . This gives us C -indexed families of maps of $A \rightarrow B$.

The important case

We **really** care about the case when $B = \boxed{\Omega}$ because this allows us to internalize predicates as truth-valued maps. More to come!

Subobjects

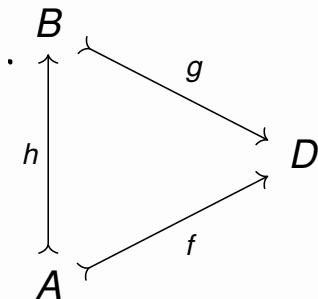
Recall

A **subobject** of $d \in \mathcal{C}$ is a monomorphism $f : a \rightarrowtail d$ with codomain d .

Subsets

$$A \subseteq B$$

$$\forall a \in A, a \in B$$



$$A = \{1, 2\}$$
$$h:$$
$$B = \{3, 4\}$$

Hand-drawn mapping from $A = \{1, 2\}$ to $B = \{3, 4\}$. An arrow from 1 to 3 and an arrow from 2 to 4 are shown, with a large 'X' drawn over the arrows, indicating a non-injective or non-monomorphic mapping.

Intuition

In **Set**, note that $A \subseteq_{\text{Set}} B \subseteq_{\text{Set}} D$ may not hold. However, $f(A) = g(h(A)) \subseteq_{\text{Set}} g(B) \subseteq_{\text{Set}} D$ does hold.



Structure of Subobjects

Intuition

Subset ($\subseteq_{\text{Set}}$) forms a **partial ordering**, being reflexive, transitive, and antisymmetric. However, $f \subseteq g$ and $g \subseteq f$ only implies f and g are **isomorphic**, not equal. So $f \simeq g$, not $f = g$.

We really want subobjects to behave like subsets, so we replace f with its equivalence class $[f]$ so that we gain equality (and thus antisymmetry). So, $\subseteq$ with **equivalence classes** of subobjects is a **partial order**.

Note

In practice, we usually refer to subobjects as the equivalence classes of such monomorphisms (i.e. modulo renaming). We drop the brackets around $[f]$ to f .

$[f]$

Elements and Naming Arrows

$$a \in B$$

Definition

Elements are determined as subobjects from a terminal object 1 .

$$f : 1 \longrightarrow A$$

$$f(0) = a \in A$$

Definition

A **naming arrow** ($\ulcorner f \urcorner$) is an element of an exponential that allows you to talk about morphisms internal to the category.

$$f : 1 \longrightarrow B^A$$

Subobject Classifier

A **subobject classifier** $\Omega \in \mathcal{C}$ is an object with a morphism that satisfies the Ω -axiom.

Ω -Axiom

For each monomorphism $f : a \rightarrowtail b$, there exists a unique morphism $\chi_f : b \rightarrow \Omega$ such that this diagram is a pullback

$$\begin{array}{ccc} a & \xrightarrow{f} & b \\ \downarrow ! & \lrcorner & \downarrow \chi_f \\ 1 & \xrightarrow{\text{true}} & \Omega \end{array}$$

$$\chi_f(a) = \sum_{a \in A} 1$$

Topos Definition and Examples

Definition

Definition

An **elementary topos** is a category $\mathcal{E}$ such that

- ▶ $\mathcal{E}$ is finitely complete
- ✓ $\mathcal{E}$ is finitely cocomplete
- ▶ $\mathcal{E}$ has exponentials
- ▶ $\mathcal{E}$ has a subobject classifier

Examples

► **Set**

- Motivation for elementary topoi

► **Finset**

- Inherits topos structure from **Set**

► **Finord**

- Every finite set is isomorphic to a finite ordinal, so the topos properties transfer from **Finset**

► **Set**²

- The category of pairs of sets

$\langle A, B \rangle$

► **Set**[→]

- The category of functions

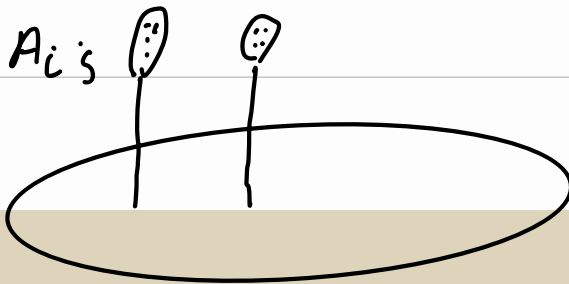
► **M-Set**

- A pair (X, λ) where X is a set and λ is a monoid action.

$M \quad X$

Sheaves and Monoids

Sheaves as Local Sets



Intuition

Sheaves are like sets varying over a space

Sheaf theory is very important, but the details of bundles and sheaves will be omitted here as we do not have the necessary background yet.

The takeaway should be that the definition of a topos allows us to **localize truth** using sheaves. So, instead of truth values being fixed, they become open regions subject to varying conditions.

Modal logic $\rightarrow$ Kripke Joyal

Monoids

Definition

A monoid is a set M with an action $*$: $M \times M \rightarrow M$ and a distinguished element e such that:

1 $\forall m, p, q \in M, (m * p) * q = m * (p * q)$ — assoc.

2 $\forall m \in M, e * m = m = m * e$ — id

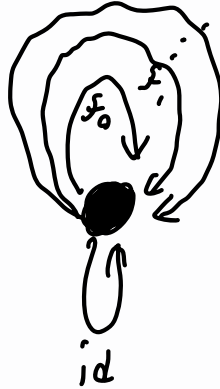
Intuition

Monoids are just like groups, but without requiring inverses.



Monoids as Categories

Assoc.
id.



Actions

Definition

An **action** of M on X is a function $\lambda : M \times X \rightarrow X$ defined as $(\forall m \in M)(\forall x \in X), \lambda(m, x) = \lambda_m(x)$ with λ having analogues of associativity and unitality.

$$\lambda(m, x) = m \cdot x \neq m * x$$

Intuition

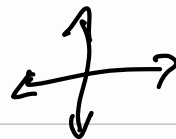
The constraints the monoid imposes on the action are:

I. $e \cdot x = x$ ↙

II. $m \cdot (p \cdot x) = (m * p) \cdot x$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ M & X & M & X \end{array}$$

M-sets



Definition $(X, \cdot)$

An **M-set** is a pair (X, λ) with $\lambda : M \times X \rightarrow X$ an action of M on X

Example

X is the set of points in the Euclidean plane and M is the group of Euclidean transformations with $*$ function composition. The M -action on X is $m(x)$, applying a transformation m to x .

The identity in this case is the identity transformation with $\text{id}(x) = x$

For associativity, we see that $m \cdot (p \cdot x) = m(p(x)) = (m \circ p)(x) = (m * p) \cdot x$.

Set theory analogues

Overview

► **Finite completeness** gives **finite set manipulation**

- Terminal object $\approx$ singleton set
- Categorical product $\approx$ cartesian product
- Pullbacks $\approx$ preimages or compatible pairs
- Equalizers $\approx$ solutions to equations

} - 1 lecture

► **Finite cocompleteness**

- Initial object $\approx \emptyset$
- Coproduct $\approx$ disjoint union
- Pushouts $\approx$ gluing
- Coequalizers $\approx$ quotient sets

} 1 lecture

► **Exponentials** $\approx$ internal function spaces

- Power objects $\approx$ power sets

} - 1 lecture

► **Subobject classifier** (Ω) $\approx$ internal truth values and predicates

- ZF Comprehension (Separation) Principle $\approx \Omega$ -axiom

} 1 lecture

Power Sets

$$\mathcal{P}(A) = \{ U \subseteq A \}$$

We often think of the power set $\mathcal{P}(A)$ as the set of subsets of A . However, the **role** power sets play in **Set** is much more important when we define the more general **power object**.

Intuition

Relations $R \subseteq B \times A$ correspond to functions $B \rightarrow \mathcal{P}(A) \cong 2^A$.

For each element of B , we get a subset of A via $R_b = \{ a \in A : (b, a) \in R \}$. We also get a corresponding function $f_R : B \rightarrow \mathcal{P}(A)$ with $f_R(b) := R_b$.

Power Set Example

Consider $R := x < y$ for $x \in X = \{1, 2\}$ and $y \in Y = \{1, 2, 3, 4\}$.

Recall

For each element of X , we get a subset of Y via $R_x = \{y \in Y : (x, y) \in R\}$. We also get a corresponding function $f_R : X \rightarrow \mathcal{P}(Y)$ with $f_R(x) := R_x$

$$f_R(1) = \{2, 3, 4\}$$

$$\phi = 0$$

$$\{\phi\} = 1$$

$$\{1, 0\} = 2$$

Power Object Definition

Definition

A category $\mathcal{C}$ with products has power objects if for any $\mathcal{C}$ -object a , there exists $\mathcal{C}$ -objects $\mathcal{P}(a)$ and $\in_a$ and a monomorphism $\in : \in_a \rightarrow \mathcal{P}(a) \times a$ such that for any $\mathcal{C}$ -object b , and "relation" $r : R \rightarrow b \times a$ there is exactly one $\mathcal{C}$ -arrow $f_r : b \rightarrow \mathcal{P}(a)$ with a pullback in $\mathcal{C}$ of the form:

$$\in_a = \{ \langle U, \pi \rangle : \\ U \subseteq A, \pi \in U \}$$

$$\begin{array}{ccc} R & \xrightarrow{r} & b \times a \\ \downarrow ! & \lrcorner & \downarrow f_r \times \text{id}_a \\ \in_a & \xrightarrow{\in} & \mathcal{P}(a) \times a \end{array}$$

Huh?

Let us reexamine our case when, for each element of B , we get a subset of A via $R_b = \{a \in A : (b, a) \in R\}$. We also get a corresponding function $f_R : B \rightarrow \mathcal{P}(A)$ with $f_R(b) := R_b$.

We note that $\in_A := \{\langle U, a \rangle : U \subseteq A \text{ and } a \in U\}$.

$$\begin{array}{ccccc} B \times A & \supseteq & R & \xrightarrow{r} & B \times A \\ \downarrow \scriptstyle \langle \cdot, a \rangle & & \downarrow \scriptstyle ! & \lrcorner & \downarrow \scriptstyle f_R \times \text{id}_A \\ \langle R_b, a \rangle & & \in_A & \xrightarrow{\in} & \mathcal{P}(A) \times A \end{array}$$

Every Topos has Power Objects

Theorem

Any topos $\mathcal{E}$ has power objects

Sketch of proof: For some object a , take $\mathcal{P}(a) = \Omega^a$ and $\in: \in_a \rightarrow \Omega^a \times a$ be the subobject with character ev_a . From the Ω -axiom, we have:

$$\begin{array}{ccc} \in_a & \xrightarrow{\in} & \Omega^a \times a \\ \downarrow ! & \lrcorner & \downarrow \text{ev}_a \\ 1 & \xrightarrow{\text{true}} & \Omega \end{array}$$

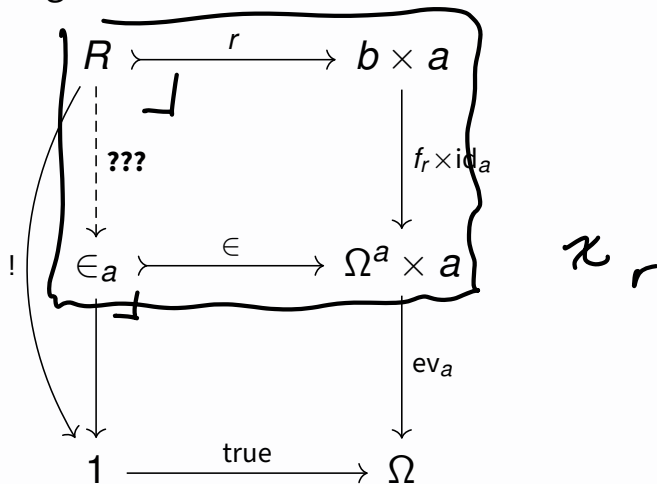
Proof Cont.

Since topoi have exponentials, we have:

$$\begin{array}{ccc} & \Omega \times a & \\ & \uparrow f_r \times \text{id}_a & \searrow \text{ev}_a \\ b \times a & & \Omega \\ & \nearrow \chi_r & \end{array}$$

Proof Cont.

Now we squish the diagrams together:



Since $\text{ev}_a \circ (f_r \times \text{id}_a) = \chi_r$, the perimeter is a pullback by the Ω -axiom, so our dubious morphism does exist and is unique. In fact, every square is a pullback by the two pullback lemma.

Conclusion of Proof

So, we have the following diagram as desired:

$$\begin{array}{ccc} R & \xrightarrow{r} & b \times a \\ \downarrow & \lrcorner & \downarrow f_r \times \text{id}_a \\ 1 & \xrightarrow{\epsilon} & \mathcal{P}(a) \times a \end{array}$$

Comprehension

$$\{x: \varphi(x) = T\}$$

Recall

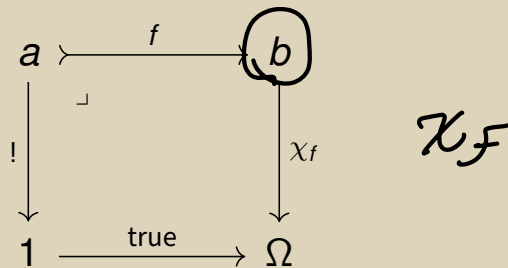
Recall, the **ZF Separation Principle** tells us for a set A and condition $\varphi(x)$,
 $\exists \{x : \boxed{x \in A} \text{ and } \varphi(x)\}.$

Usually, we think of this in terms of **set ontology**. However, the subobject classifier tells us subobjects determine predicates.

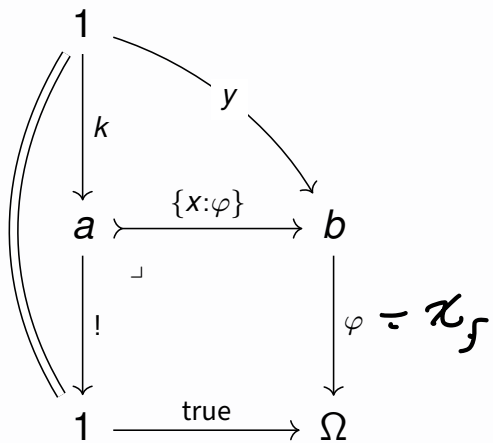
Ω -Axiom

Recall

Ω -Axiom: For each monomorphism $f : a \rightarrowtail b$, there exists a unique morphism $\chi_f : b \rightarrow \Omega$ such that this diagram is a pullback



Comprehension Diagrams



Topos Structure

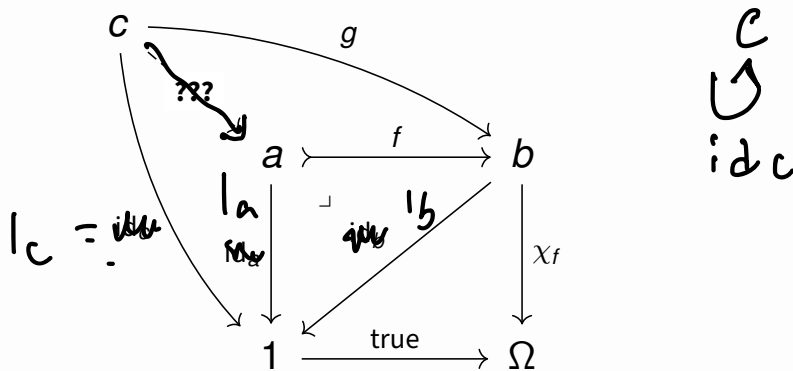
Monics Equalize

Theorem

In a topos $\mathcal{E}$, if $f : a \rightarrowtail b$ is a monomorphism, then f is an equalizer of χ_f and $\text{true}_b = \text{true} \circ \text{id}_b$

Proof

We use the universal property of pullbacks to show the following dubious morphism must exist:



Since the outer perimeter commutes, our dubious morphism must exist.

Epi-Mono Factorization of Set

Consider a function $f : A \rightarrow B$. We note $|A| \overset{\geq}{\underset{>}{\geq}} |f(A)|$, so if we restrict the codomain to just the image, we get a surjective function equal to f . The image $f(A) \subseteq B$ and any function $g : f(A) \rightarrow B$ is injective.

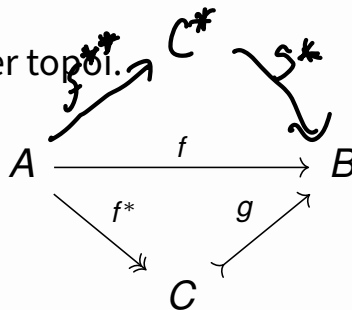
$$|A| \geq |f(A)|$$

$$f : A \rightarrow B$$

$$f : A \xrightarrow{\sim} f(A) \xrightarrow{\hookrightarrow} B$$

Epi-Mono Factorization

This situation generalizes in other topoi.



$$A = \{0, 1\}$$

$$B = \{0, 1, 2\}$$

$$A = f(A)$$

$$A: \{0, 1\}$$

$$\{0, 1\}$$

$$C = \{*, 1\}$$

$$f(A) = \{0, 1\}$$

In fact, it is unique up to unique commuting isomorphism.

Various Consequences of the FTOT

Fact 1

Pullbacks preserve epimorphisms in a topos.

For a pullback square in a topos:

$$\begin{array}{ccc} a & \longrightarrow & b \\ \downarrow g & \lrcorner & \downarrow f \\ c & \longrightarrow & d \end{array}$$

if f is an epimorphism, then g is an epimorphism.

Various Consequences of the FTOT

Fact 2

Coproducts preserve pullbacks in a topos.

$$\begin{array}{ccc}
 a & \xrightarrow{f} & d \\
 \downarrow \scriptstyle{g} \lrcorner & & \downarrow \scriptstyle{k} \\
 b & \xrightarrow{h} & e
 \end{array}
 \quad \text{and} \quad
 \begin{array}{ccc}
 a' & \xrightarrow{f'} & d' \\
 \downarrow \scriptstyle{g'} \lrcorner & & \downarrow \scriptstyle{k'} \\
 b' & \xrightarrow{h'} & e'
 \end{array}$$

$$\begin{array}{ccc}
 a + a' & \xrightarrow{[f, f']} & d \\
 \downarrow \scriptstyle{g + g'} \lrcorner & & \downarrow \scriptstyle{k} \\
 b + b' & \xrightarrow{[h, h']} & e
 \end{array}$$

Exercises

Power Objects in Various Topoi

Recall, a category $\mathcal{C}$ with products has power objects if for any $\mathcal{C}$ -object a , there exists $\mathcal{C}$ -objects $\mathcal{P}(a)$ and $\in_a$ and a monomorphism $\in : \in_a \rightarrow \mathcal{P}(a) \times a$ such that for any $\mathcal{C}$ -object b , and "relation" $r : R \rightarrow b \times a$ there is exactly one $\mathcal{C}$ -arrow $f_r : b \rightarrow \mathcal{P}(a)$ with a pullback in $\mathcal{C}$ of the form:

$$\begin{array}{ccc} R & \xrightarrow{r} & b \times a \\ \downarrow & \lrcorner & \downarrow f_r \times \text{id}_a \\ \in_a & \xrightarrow{\in} & \mathcal{P}(a) \times a \end{array}$$

Table of Power Object Components

$$B^A \quad A = \{2, 4\} \quad B = \{0, 1\} \quad 2^2 = 4$$

Topos	Object	Arrow	Ω	Y^X
Set	set	function	2	hom-set
Finset	finite set	function	2	hom-set
Finord	finite ordinal	function	2	cardinality of hom-set
Set ²	pair of sets	pair of functions	(2, 2)	hom-set $\times$ hom-set
Set ^{$\rightarrow$}	function $X_0 \rightarrow X_1$	commutative square	$0 \mapsto 0$ $\frac{1}{2}, 1 \mapsto 1$	{comm. squares} $\rightarrow Y_1^{X_1}$
M-Set	monoid action	equivariant map	(left ideals L_M , "right division" ω)	(E, σ) s.t. E is hom-set of $(M, *) \times (X, \lambda) \xrightarrow{f} (Y, \mu),$ $(n \cdot_\sigma f)(m, x) = f(m * n, x)$

Example: Set

$$\mathcal{P}(a) \text{ and } \epsilon_a \qquad \epsilon : \epsilon_a \rightarrow \mathcal{P}(A) \times a$$

$$r : R \rightarrow b \times a \qquad f_r : b \rightarrow \mathcal{P}(A)$$

$$\begin{array}{ccc} R & \xrightarrow{\quad} & b \times a \\ \downarrow & \lrcorner & \downarrow f_r \times \text{id}_a \\ \epsilon_a & \xrightarrow{\quad} & \mathcal{P}(A) \times a \end{array}$$

$\mathcal{P}(A)$ = power set

Example: Finset

$$|\mathcal{P}(A)| < \infty$$

$$2^n \approx \mathcal{P}(A)$$

Example: Finord

$$|2^n| \cong \mathcal{P}(A)$$

$$\begin{array}{ccc} R & \xrightarrow{\kappa} & b \times a \\ \downarrow \dashv & & \downarrow \pi \times \text{id}_c \\ \mathcal{C}_a & \xrightarrow{\quad} & \mathcal{P}(A) \times a \end{array}$$

Example: Set²

Example: Set[→]

Table of Power Objects

Topos	Object X	$\mathcal{P}(X)$
Set	X	2^X
Finset	X	2^X
Finord	n	$ 2^n $
Set²	(X_1, X_2)	$2^{X_1} \times 2^{X_2}$
Set[→]	$\begin{array}{c} X \\ f \downarrow \\ Y \end{array}$	$\left\{ \begin{array}{ccc} X & \xrightarrow{h} & \{0, \frac{1}{2}, 1\} \\ f \downarrow & & \downarrow t \\ Y & \xrightarrow{k} & 2 \end{array} \right\} \rightarrow 2^Y$
M-Set	(X, λ)	(E, σ) s.t. E is hom-set of $(M, *) \times (X, \lambda) \xrightarrow{f} (L_M, \omega),$ $(n \cdot_\sigma f)(m, x) = f(m * n, x)$

Set^2

objects (X, Y)

Subobject (S, T) where $S \subseteq X, T \subseteq Y$
 $\chi_S: X \rightarrow 2 \quad \chi_T: Y \rightarrow 2$

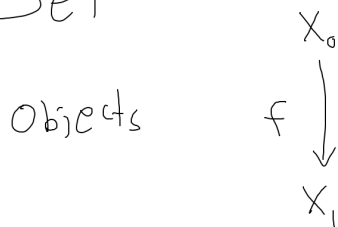
$$\Omega = (2, 2)$$

$$(X, Y)^{(U, V)} = (X^U, Y^V)$$

$$\therefore \text{Power object of } (X, Y) \quad \Omega^{(X, Y)} = (2^X, 2^Y)$$



Set $\rightarrow$



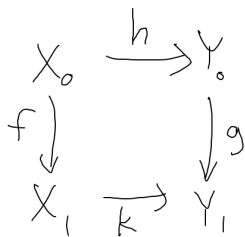
If $x \in X_0$, then

1. is $x \in S_0$?

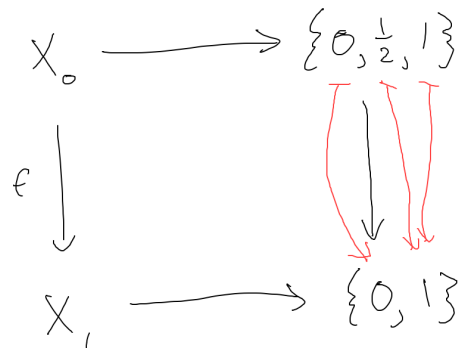
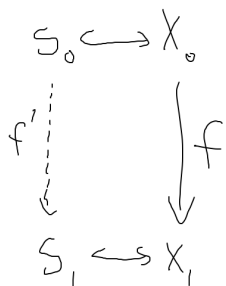
$\frac{1}{2}$. if not, is $f(x) \in S_1$?

0. otherwise, neither

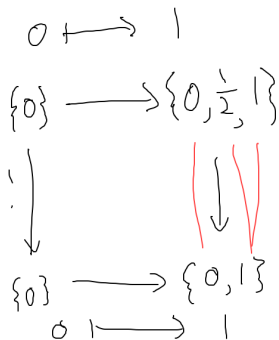
Arrows from f to g



Subobject of f ?



true



Set $\rightarrow$

$$\begin{array}{ccc}
 a & \xrightarrow{h} & h(a) \\
 \cap & & \\
 A & \xrightarrow{h} & C \\
 f \downarrow & & \downarrow g \\
 B & \xrightarrow{k} & D
 \end{array}$$

What is g^f

$$\begin{array}{ccc}
 (\langle h, k \rangle, a) & \xrightarrow{f} & h(a) \\
 \cap & & \cap \\
 \text{Hom}_{\text{Set}}(f, g) \times A & \xrightarrow{\quad} & C \\
 \downarrow g^f \times f & & \downarrow g \\
 D^B \times B & \xrightarrow{\text{ev}} & D \quad g(h(a)) \\
 & & \parallel \\
 (k, f(a)) & \xrightarrow{\quad} & k(f(a))
 \end{array}$$



Set $\rightarrow$

Given

$$\begin{array}{c} A \\ \downarrow f \\ B \end{array}$$

,

then

$$\begin{array}{c} \text{Hom}_{\text{Set}}(f, \Omega) \\ \downarrow \Omega^f \\ 2^B \end{array}$$



M-Set

$$(M, *) \in \text{M-Set}$$

Objects (X, λ) where $\lambda: M \times X \rightarrow X$ assoc.
unital

Arrow $f: (X, \lambda) \rightarrow (Y, \mu)$ is an $f: X \rightarrow Y$ st,

$$\forall m \in M \quad \begin{array}{ccc} X & \xrightarrow{\lambda_m} & m \cdot_\lambda X \\ \downarrow f & & \downarrow f \\ f(X) & \xrightarrow{\mu_m} & m \cdot_\mu f(X) \end{array}$$

"equivariant"

(X, λ)
 (Y, μ)

~~$(Y, ?)$~~

~~$(\text{Hom}_{\text{M-Set}}((X, \lambda), (Y, \mu)), ?)$~~

$$\left(\text{Hom}_{\text{M-Set}}((M, *) \times (X, \lambda), (Y, \mu)), \sigma \right) \quad \left(\underset{M}{n} \cdot \underset{\text{Hom}(\dots)}{\sigma} f \right) \left(\underset{M}{m}, \underset{X}{x} \right) = f(m * n, x)$$



M-Set

Given $B \subseteq M$ and $m \in M$,

$$\Omega = (L_M, \omega)$$

$$B : m := \{n \in M : n * m \in B\}$$

$$B \in L_M \iff (\forall m \in M) (\forall b \in B) (m * b \in B)$$

i.e. $MB \subseteq B$

$$B \in L_M \implies B : m \in L_M$$

$$m \cdot_\omega B := B : m$$

Power object for (X, λ)

Ω

$$\left(\text{Hom}_{M\text{-set}} \left((M, *) \times (X, \lambda), (L_M, \omega) \right), \sigma \right)$$



Reading for Next Lecture

- ▶ Minimal
 - Section 5 of chapter 5
 - Sections 1 and 5-7 of chapter 6
- ▶ Excellent
 - Sections 4 and 5 of chapter 5
 - All of chapter 6
- ▶ Extra
 - 12-26 videos of this playlist